AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM112

Introduction to Computing and Information Technology

Midterm Examination (MTA)
Fall Semester 2025/2026

Date: 20/Nov./2025

nber of Exam Pages: 2 uding this cover sheet)	**Time Allowed:** 2 hours

tions:

- Total Marks: **30**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

CamScanner

Part I: Multiple Choice Questions (5 marks):

1) A Which of the following is a web browser?
   a) Google Chrome
   b) Gmail
   c) Facebook
   d) YouTube
2) Binary number 00001010 is equal to:
   a) 5
   b) 10
   c) 8
   d) 15
3) Ubiquitous means:
   a) Unavailable around us
   b) Everywhere around us
   c) High cost
   d) Low cost
4) To represent 256 unique values in digital color coding, we need:
   a) 8 bits
   b) 4 bits
   c) 16 bits
   d) 2 bytes
5) ----------refers to the ability of a UI to prevent errors if possible, or to make them easy to recover from, if not.
   a) Visibility
   b) Tolerance
   c) affordance.
   d) simplicity,

Part 2: Short questions (10 marks):

Question 1: (3 marks)
List two solutions to reduce the amount of memory required to store an image

Question 2: (4 marks)
For the given numbers find their equivalent in the specified base. **Show your steps.**

$(101101)_2 = (?)_8$ $(645)_8 = (?)_{16}$

$(2F)_{16} = (?)_{10}$ $(3.25 \times 4)_{10} = (?)_2$

Question 3: (3 Marks)
In a sound wave, Give the definition of **Cycle**, **Frequency** and **Amplitude**?

Part 3: Problem solving / Long Questions (15 marks)

Question 1: (6 marks)
A 2-hour surveillance video takes up 512 GB of storage. If each frame uses 2 MB, what is the frame rate (fps)?

Question 2: (4 marks)
List two techniques used in lossless compression and describe each one of them

Question 3: (5 marks)

a) What Is Data Science?
b) List 4 Skills required to Become a Data Scientist?

End of question

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